

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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I. WORK DONE THIS WEEK:-

In the second week, we started implementing the initial steps from our project ideas. We started off with looking into fabric dataset and studying some of the sample images. In the process, we tried to gain an understanding of what the images look like and the differences between the defect and normal fabric texture. It was necessary since the problem is not just a question of finding some object in an image but the whole surface texture. The fabric surface consists of the same repeating pattern and so a defect might be a relatively small change in the complicated texture.

We started off by examining the images in their original form and applying some basic preprocessing to some sample images. First, we tried converting the images into grayscale since our goal is the texture analysis. Therefore, grayscale images are simpler for the initial examination and we can work only with the intensity information. We compared the original images with the converted ones and made sure that all the necessary texture patterns were present.

At the same time, we tried to examine whether any additional preprocessing of the images, such as filtering and resizing, was going to be necessary for our dataset. At this stage, we did not want to use any additional preprocessing, since even small changes in the texture might contain the important information regarding the defect. For now, we keep it simple and try to

figure out the effects of every step before using it.

After implementing the basic preprocessing steps, we started implementing the initial part of the algorithm for the texture analysis. We concentrated mainly on Gray-Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP) since they were the two main descriptors that we had chosen last week. We tried to figure out the parameters used for the calculation of these descriptors and the effect of their changes on the output.

With respect to GLCM, we tried to figure out the changes in the relation of the neighboring pixels depending on the directions and distances used. Moreover, we started to extract the basic texture features such as contrast, homogeneity, energy and correlation from the sample regions. It was not the goal to detect the defect at this stage, but to figure out the information in numerical form that we get using this descriptor.

At the same time, we tried to experiment with LBP applied to some sample regions. It was done in order to figure out how the local patterns in the texture are described in LBP form. Also, we wanted to understand if the defect and the normal regions are presented differently in the LBP representation. We are still figuring out the best parameters to use in our final algorithm.

Another aspect that we started thinking about this week is the size of the region used for the texture analysis. Since our project involves the localization task, we cannot analyze just the whole image and find one texture value. We will need to analyze either regions or windows in order to localize the defect. However, using either too big or too small window might affect the result, so we are trying to understand it first.

As of now, our implementation is not complete and involves only the first steps of the pipeline. We did not implement the adaptive defect detection and localization algorithms yet. Our implementation looks like:

Fabric Image → Basic Preprocessing → Initial Texture Extraction → Further Analysis

So, the work of this week mostly helped us connect our ideas of Week 1 with real experiments with the images.

II. WORK TO BE DONE NEXT WEEK:-

During the next week, we will continue working on texture descriptors. Our aim is to test those on a larger set of fabric images and compare texture information received from normal and defective regions of the images. It will give us a chance to investigate whether there is a difference between those regions or not.

We will continue our work with GLCM and LBP parameter settings. Our aim is to analyze how the distance, direction and region size affect the resulting texture feature. It is important not to define final parameter values yet since they might act differently for other fabrics.

It will also be important to work with the concept of splitting image into smaller windows. Several small window sizes will be tried and analyzed for their influence on texture feature changing from one region to another. This will help us to investigate how the local texture information can be further used for defect localization.

In addition, we will continue the investigation of the preprocessing stage. Filtering techniques will be tested if needed and their results will be compared with the original image. It will help us to understand whether the preprocessing stage helps in improving the texture analysis without erasing valuable information about the fabric.

Along with the implementation, we will continue the reading of relevant papers about GLCM,

LBP and fabric defect detection.

III. LITERATURE REVIEW:-

In this week, we proceeded with the literature review trying to understand the practical application of texture descriptors. Since we have started developing GLCM and LBP implementations, we have revisited the articles reviewed earlier and tried to relate them to our current experiments.

The article "Fabric Defect Detection Using Modified Local Binary Patterns" was quite helpful in understanding the application of local binary patterns in fabric defect detection. The very idea of describing local patterns in textures is applicable to our project, since a defect produces a local pattern which differs from the texture of the rest of the fabric.

The article "Fabric Defect Detection based on GLCM Approach" provided us with a good insight into the role of statistical texture features extracted using the Gray-Level Co-occurrence Matrix. Such features as contrast, homogeneity, energy and correlation offer several different approaches to describe the texture of an image region.

One more paper we have reviewed was "Fabric Defect Classification Based on LBP and GLCM". It was the most relevant to our current investigation since it uses both local texture features described by LBP and statistical texture features described by GLCM. At this point of the project, we are investigating the two methods independently in order to study the behaviour of each of them before considering whether their features could be combined.

At the same time, the literature review made us understand that the parameters of texture description are important for defect detection. The distance and direction used for GLCM and the neighbourhood used for LBP influence the features extracted from the image. That is one of the reasons why at this stage of the work we experiment with the initial settings of the methods instead of defining their final configuration right away.

We have not yet managed to find out how to adapt the descriptors for defect detection. That is why in our further literature review we will try to concentrate on methods of local texture comparison and defect decision making.

IV. CHALLENGES FACED:-

One of the problems we encountered during this week is finding out how much preprocessing should actually be done with the fabric images. In regular image processing tasks, filters and smoothers can help improve the quality of the image. In our case, however, texture information is what really matters. Excessive use of the preprocessing techniques might lead to loss of important information.

The other problem we encountered is understanding the impact of the GLCM parameters on the output. Changing the distance or direction used for GLCM calculation can give quite different features of the texture. So, we need to do some experimenting to select the appropriate parameters for further use in the project.

Another problem is that choosing the size of the local area is not a simple task. Too small an area might not provide enough information regarding the whole texture of the fabric. On the contrary, too big an area might hide a small defect behind the rest of the normal texture. As localization is one of our goals in this project, we need to consider this issue.

It is still quite difficult to determine whether a certain change in the texture means the presence of a defect or normal variance in the fabric. So, we have not yet begun with the detection stage.

We need to understand the behavior of our texture descriptors first.

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